

PID controlled L-DOPA model for automated drug delivery



EE66103 Minor Project

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Abstract

Parkinson's Disease (PD) is a complex and debilitating neurological disorder affecting millions worldwide. This project focuses on the hardware synthesis of a closed-loop Proportional-Integral-Derivative (PID) controlled Levodopa (L-DOPA) infusion system for automated drug delivery in Parkinson's disease patients using AD844 Current Feedback Operational Amplifiers (CFOAs). The Duodopa pump, which delivers L-DOPA gel to the small intestine, offers advantages like improved symptom control and motor function. However, manual control poses several challenges leading to inadequate relief or side effects.

To address these challenges, a PID controller is proposed to regulate the plasma L-DOPA concentration. The PID controller is implemented using CFOAs, which offer advantages over traditional voltage feedback op-amps including high slew rate, constant bandwidth, and better performance in integrator and differentiator circuits. The closed-loop system is synthesized using AD844 CFOAs, resistors, and capacitors. The L-DOPA pharmacokinetic model, represented by a second-order system with delay, is designed and synthesized using first-order low-pass filters and a Sallen-Key filter.

The hardware-synthesized system effectively tracks the given setpoint signal, with simulation results validated through PSpice software. This approach offers a cost-effective and reliable solution for continuous drug delivery, potentially improving the quality of life for patients with advanced Parkinson's disease.

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Abbreviations

AAAD	Aromatic L-Amino Acid Decarboxylase
BG	Basal Ganglia
BBB	Blood-Brain Barrier
CFOA	Current Feedback Operational Amplifier
CNS	Central Nervous System
DAQ	Data Acquisition System
DBS	Deep Brain Stimulation
LCIG	Levodopa/Carbidopa Intestinal Gel
L-DOPA	Levodopa
Op-amp	Operational Amplifier
PD	Parkinson's Disease
PEG-J	Percutaneous Endoscopic Gastrojejunostomy
PID	Proportional-Integral-Derivative
PK	Pharmacokinetics
PK-PA	Pharmacokinetics-Pharmacological Activation

Chapter 1

Introduction to Automated Drug Delivery for Parkinson's Disease

1.1 Overview of Parkinson's Disease

Parkinson's Disease (PD) is a complex and debilitating neurological disorder that affects millions of people worldwide. It is a lifelong, idiopathic, chronic neurological condition that progressively destroys nerve cells in the central nervous system (CNS), especially in the basal ganglia region of the brain. A variety of complex brain circuits comprising many loops and inhibitory networks are damaged by Parkinson's disease, leading to various motor and non-motor symptoms associated with aging and the chronic nature of the disease.

When the brain's ability to produce dopamine drops or stops, little or no signal is delivered, weakening one's ability to regulate their movements. When dopamine levels fall by 80%, this results in motor symptoms [1]. Pharmacological treatments to restore striatal dopaminergic neurotransmission may temporarily reverse Parkinsonism.

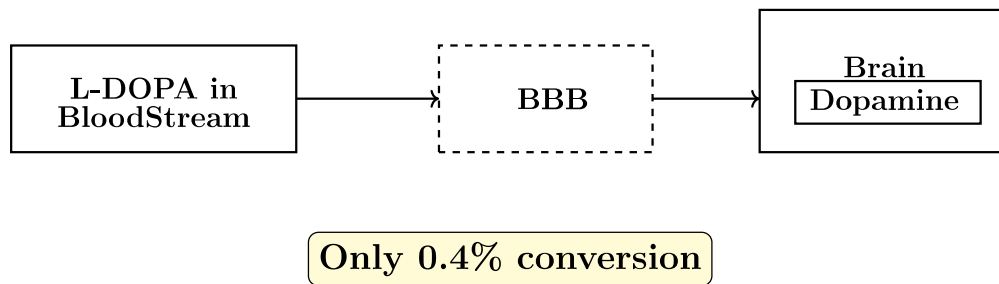


Figure 1.1: L-DOPA transport across blood-brain barrier and conversion to dopamine.

1.2 L-DOPA and Dopamine Pathway

Dopamine itself cannot pass the blood-brain barrier (BBB), making direct dopamine delivery ineffective. Levodopa (L-DOPA), a naturally occurring amino acid, serves as an intermediate stage in dopamine synthesis. Levodopa is actively transported from the upper small intestine into the bloodstream upon oral administration through a mechanism specific for large, neutral L-amino acids.

Once in the brain, aromatic L-amino acid decarboxylase (AAAD) quickly converts Levodopa into dopamine. Levodopa's pharmacologic impact is mostly caused by conversion to dopamine, but it also directly affects neurotransmitters and neuromodulators,

which may contribute to its antiparkinsonian outcome [2].

1.3 The Duodopa Pump System

The Duodopa pump delivers Levodopa/Carbidopa intestinal gel (LCIG) continuously into the small intestine, providing a consistent and stable daily dose of medication. This advanced delivery method offers several clinical benefits:

- **Improved symptom control** — Steady drug levels reduce motor fluctuations
- **Fewer medication fluctuations** — Eliminates the "peak-and-trough" effect of oral pills
- **Enhanced motor performance** — Patients experience more predictable mobility
- **Better quality of life** — Reduced "off" periods and improved daily function

The therapy utilizes a refined PEG-J tube (percutaneous endoscopic gastrojejunostomy) system that delivers LCIG directly to the proximal small intestine, where optimal absorption occurs.

1.3.1 Challenges with Manual Control

Despite its advantages, manually controlled Duodopa pumps present several clinical challenges:

- **Constant adjustments required** — Settings need frequent modification throughout the day
- **Difficult dose optimization** — Finding the ideal infusion rate is complex and time-consuming
- **Extensive training needed** — Patients and caregivers require regular education and supervision
- **Risk of human error** — Incorrect adjustments can lead to:
 - Insufficient symptom relief (underdosing)
 - Adverse effects from overdosing
 - Unpredictable "on-off" fluctuations

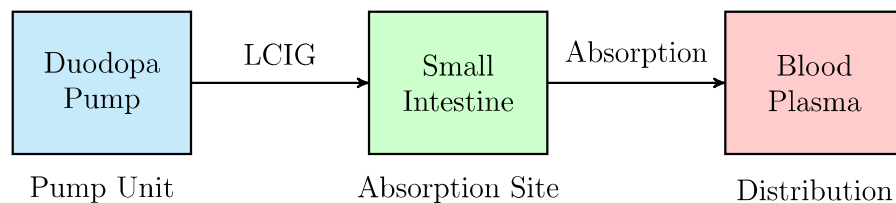


Figure 1.2: Block diagram of Duodopa infusion system

1.4 Need for Automated Drug Delivery

The limitations of manual control necessitate an automated closed-loop system that can:

1. Continuously monitor plasma L-DOPA concentration
2. Automatically adjust infusion rate based on feedback
3. Maintain therapeutic drug levels within optimal range
4. Reduce side effects and improve patient outcomes

This project addresses these needs by developing a hardware-synthesized PID controller using AD844 CFOAs for automated L-DOPA delivery.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews the existing literature on Parkinson's disease treatment, L-DOPA pharmacokinetics, and closed-loop drug delivery systems. The review covers both fundamental concepts and recent advances in the field.

2.2 Parkinson's Disease Treatment Options

Three kinds of treatment may be implemented to manage Parkinson's disease [3]:

1. Therapeutic or regenerative treatment
2. Symptomatic treatment
3. Protective or preventive treatment

"Wearing off" typically occurs three to four hours following a dosage of Levodopa, marked by the return of parkinsonian symptoms when the drug's effects start to wane. Stalevo (Levodopa, carbidopa, and entacapone combination) has been licensed to treat patients with end-of-dose "wearing-off" [4].

2.3 L-DOPA Pharmacokinetics

The pharmacokinetics of Levodopa/Carbidopa combination in individuals with advanced Parkinson's disease has been studied following two different types of administration [5]:

- Oral sustained-release
- Continuous Levodopa/Carbidopa intestinal gel infusion

The continuous and gradual infusion of Levodopa represents a significant treatment option for advanced Parkinson's disease, greatly enhancing motor fluctuation control.

Table 2.1: Comparison of L-DOPA Administration Methods

Parameter	Oral Administration	Duodenal Infusion
Absorption Pattern	Pulsatile	Continuous
Plasma Concentration	Fluctuating	Stable
Motor Fluctuations	Common	Reduced
Dosing Frequency	Multiple times/day	Continuous
Side Effects	More frequent	Reduced

2.4 Closed-Loop Drug Delivery Systems

The medical sector is increasingly adopting control systems technology. Notable advances include:

- Automated drug delivery system for focal epilepsy [6]
- Intelligent automated drug administration and therapy [7]
- Adaptive deep brain stimulation strategies [8]

2.5 PID Control in Biomedical Applications

Numerous electronic components can replicate the functionality of physical systems. Key developments include:

- Analog PID control system for quadcopter altitude control [9]
- Dynamic error correction in auto-balancing bridges [10]
- Adaptive PID control for 3D bioprinting [11]

2.6 CFOA in Analog Circuit Design

The Current Feedback Operational Amplifier (CFOA) offers distinct advantages over traditional Voltage Feedback Amplifiers (VFAs):

Table 2.2: Comparison: CFOA vs Traditional Op-Amp

Parameter	CFOA (AD844)	Traditional Op-Amp
Slew Rate	2000 V/ μ s	0.5-20 V/ μ s
Bandwidth	Constant (independent of gain)	Gain-bandwidth product limited
Input Impedance	High (inverting input low)	Very high
Feedback Type	Current feedback	Voltage feedback
Application	High-speed, integrators	General purpose

2.7 L-DOPA Pharmacokinetic Modeling

The dose-effect relationship of L-DOPA integrates pharmacokinetics and pharmacodynamics [12]. The pharmacokinetic component characterizes the relationship between drug administration and blood plasma L-DOPA concentration, modeled as a second-order linear system with delay.

$$G_{PK}(s) = \frac{Y_{L1}(s)}{Q(s)} = \frac{K_{PK}e^{-t_{d1}s}}{(1 + s\tau_1)(1 + s\tau_2)} \quad (2.1)$$

where:

- K_{PK} = gain factor (1418)
- t_{d1} = time delay (0.2461 h)
- τ_1, τ_2 = time constants (0.05473 h, 0.6073 h)

2.8 Research Gap and Motivation

While significant progress has been made in understanding L-DOPA pharmacokinetics and developing closed-loop systems, there remains a need for:

1. Hardware-implementable solutions for clinical settings
2. Cost-effective analog controllers
3. Reliable and portable systems
4. Integration with existing Duodopa pump technology

This project addresses these gaps by synthesizing a PID controller using AD844 CFOAs for automated L-DOPA delivery.

Chapter 3

System Design and Methodology

3.1 Overview of the Proposed System

The proposed closed-loop system consists of a PID controller, PK-PA model of L-DOPA, and an electrochemical microneedle L-DOPA sensor. Figure 3.1 shows the block diagram of the system.

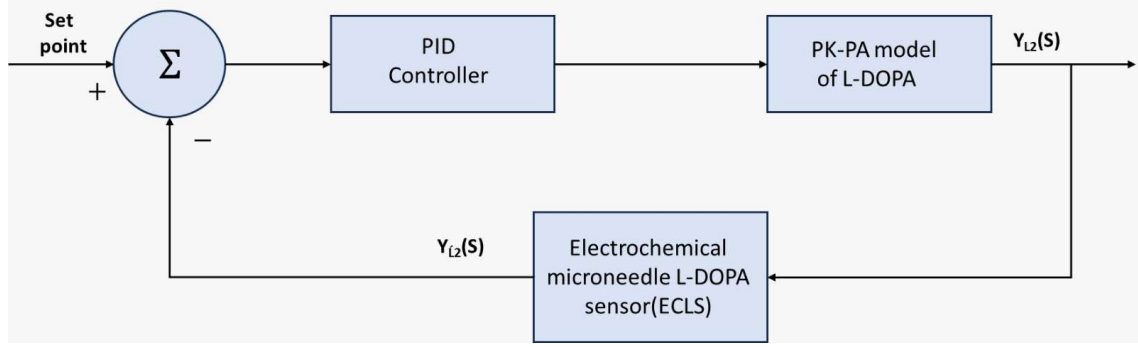


Figure 3.1: Closed loop system with PK-PA model of L-Dopa

3.2 Mathematical Modeling of L-DOPA Pharmacokinetics

The L-DOPA model integrates pharmacokinetics and pharmacological activation components. The overall transfer function is given by:

$$G_{PKA}(s) = \frac{L_d(s)}{Q(s)} = \frac{K_{PK}e^{-t_{d1}s}}{(1 + s\tau_1)(1 + s\tau_2)} \cdot Ke^{-t_{d2}s} \quad (3.1)$$

The overall gain of the plant is:

$$K_{total} = K_{PK} \times K = 1418 \times \frac{1}{250.87} = 5.6436 \quad (3.2)$$

3.3 PID Controller Design

3.3.1 PID Controller Structure

The PID controller transfer function is given by:

$$G_{PID}(s) = K_p + \frac{K_i}{s} + K_d s \quad (3.3)$$

3.3.2 Controller Tuning Method

The Hit & Trial method employed yielded the following controller gains:

Table 3.1: Controller Gains After Tuning

Tuning Method	Kp	Ki	Kd
Hit & Trial	0.1	0.4	0.044

3.4 AD844 CFOA: Architecture and Operation

The AD844 is a monolithic current feedback operational amplifier (CFOA) manufactured by Analog Devices that offers exceptional performance in integrator and differentiator circuits. Unlike traditional voltage feedback operational amplifiers, the AD844 utilizes a current feedback architecture which provides distinct advantages for high-speed analog signal processing applications.

The internal architecture of the AD844 consists of a unity-gain buffer at the input stage followed by a current mirror and an output stage. The device features two input terminals: the Y terminal (non-inverting input) which is a high-impedance voltage input, and the X terminal (inverting input) which is a low-impedance current input. This current feedback topology allows the AD844 to achieve an extremely high slew rate of $2000 \text{ V}/\mu\text{s}$ and a wide bandwidth of 60 MHz that remains constant regardless of the closed-loop gain setting.

In the PSpice implementation, the AD844 subcircuit follows the terminal assignment: Y, X, V+, V-, W, Z corresponding to pins 1, 2, 99, 50, 28, 12 respectively. The Y terminal (Pin 3) serves as the non-inverting voltage input, the X terminal (Pin 2) serves as the inverting current input, the V+ (Pin 7) and V- (Pin 4) terminals are connected to $\pm 10\text{V}$ DC supplies, the W terminal (Pin 6) provides the low-impedance voltage output, and the Z terminal (Pin 5) is a high-impedance compensation node.

Key features of the AD844 include a high slew rate of $2000 \text{ V}/\mu\text{s}$, wide bandwidth of 60 MHz, high output current of $\pm 50 \text{ mA}$, and low offset voltage of $150 \mu\text{V}$. These characteristics make the AD844 particularly suitable for integrator and differentiator circuits where fast response and high accuracy are required. The device is available in an 8-pin DIP package and is widely used in high-speed signal processing, active filters, and analog control systems.

3.4.1 PSpice Model Terminal Assignments

Based on the AD844 SPICE subcircuit definition:

<code>.SUBCKT AD844 1 2 99 50 28 12</code>
--

The terminal mapping is as follows:

Table 3.2: AD844 Terminal Mapping for PSpice Implementation

PSpice Net	Terminal	Pin	Function	In Code
1	Y	3	Non-inverting Input (Voltage)	1
2	X	2	Inverting Input (Current)	2
99	V+	7	Positive Power Supply (+10V)	40
50	V-	4	Negative Power Supply (-10V)	41
28	W	6	Output (Low Impedance)	3
12	Z	5	Compensation Node (High Impedance)	4

3.4.2 Code Example from Implementation

In the PSpice code, the AD844 is called as:

```
x1 1 2 40 41 3 4 ad844
```

This translates to:

- 1 → Y terminal (Non-inverting input)
- 2 → X terminal (Inverting input)
- 40 → V+ terminal (Positive supply: +10V DC)
- 41 → V- terminal (Negative supply: -10V DC)
- 3 → W terminal (Output)
- 4 → Z terminal (Compensation/High impedance node)

3.4.3 Key Features of AD844

- High slew rate: 2000 V/ μ s
- Wide bandwidth: 60 MHz
- High output current: ± 50 mA
- Low offset voltage: 150 μ V
- Available in 8-pin DIP package

3.4.4 CFOA Pin Configuration

Table 3.3: AD844 Physical Pin Configuration

Pin	Symbol	PSpice Terminal	Function
1	NC	-	No Connection
2	IN-	X	Inverting Input
3	IN+	Y	Non-inverting Input
4	V-	V- (50)	Negative Power Supply
5	NC/Z	Z	No Connection / Compensation
6	OUT	W	Output
7	V+	V+ (99)	Positive Power Supply
8	COMP	-	Compensation (Internal use)

3.4.5 Complete PSpice Netlist with Embedded AD844 Model

The following is the complete, self-contained PSpice implementation with the AD844 SPICE macro-model embedded:

Listing 3.1: Complete PSpice Implementation with Embedded AD844 Model

```
* PID Controller with Plant in closed loop
* Using AD844 CFOA (Current Feedback Operational Amplifier)

* =====
* AD844 SPICE Macro-model (Embedded)
* Description: 60 MHz, 2000 V/us Current Feedback Op Amp
* Node assignments:
*          (1) non-inverting input (Y)
*          (2) inverting input (X)
*          (99) positive supply (V+)
*          (50) negative supply (V-)
*          (28) output (W)
*          (12) compensation node (Z)
* =====

.SUBCKT AD844 1 2 99 50 28 12
*
* INPUT STAGE
*
R1 99 8 1E3
R2 10 50 1E3
V1 99 9 11
D1 9 8 DX
V2 11 50 11
D2 10 11 DX
I1 99 5 258E-6
I2 4 50 258E-6
Q1 50 3 5 QP
Q2 99 3 4 QN
Q3 8 6 30 QN
```

```

Q4    10  7 30  QP
R3     5   6    300E3
R4     4   7    300E3
*
* INPUT ERROR SOURCES
*
GB1  99  1    POLY(1)  1  22  150E-9  90E-9
GB2  99 30    POLY(1)  1  22  200E-9  90E-9
VOS   3   1    50E-6
LS1  30  2    1E-8
CS1  99  2    1E-12
CS2  50  2    1E-12
*
EREF 97  0    22  0  1
*
* GAIN STAGE & DOMINANT POLE
*
R5    12 97    3E6
C3    12 97    5.5E-12
G1    97 12    99  8  1E-3
G2    12 97    10 50  1E-3
V3    99 13    4.3
V4    14 50    4.3
D3    12 13    DX
D4    14 12    DX
*
* POLE AT 70 MHZ
*
R8    17 97    1E6
C4    17 97    3.18E-15
G4    97 17    12 22  1E-6
*
* POLE AT 300 MHZ
*
R12  21 97    1E6
C8    21 97    0.318E-15
G8    97 21    17 22  1E-6
*
* OUTPUT STAGE
*
ISY  99 50    5.1E-3
R13  22 99    16.7E3
R14  22 50    16.7E3
R15  27 99    30
R16  27 50    30
L2   27 28    6E-8
G9   25 50    21 27  33.33E-3
G10  26 50    27 21  33.33E-3
G11  27 99    99 21  33.33E-3
G12  50 27    21 50  33.33E-3
V5   23 27    0.5

```

```

V6  27 24      0.5
D5  21 23      DX
D6  24 21      DX
D7  99 25      DX
D8  99 26      DX
D9  50 25      DY
D10 50 26      DY
*
* MODELS USED
*
.MODEL QN      NPN(BF=1E9 IS=1E-15)
.MODEL QP      PNP(BF=1E9 IS=1E-15)
.MODEL DX      D(IS=1E-15)
.MODEL DY      D(IS=1E-15 BV=50)
.ENDS AD844

* =====
* MAIN CIRCUIT
* =====

* Power Supplies
VDD 40 0 DC 10
VSS 0 41 DC 10

* Input Signal (Step input: 0 to 1V)
VIN 1 0 PULSE (0 1 1N 1N 1n 10 20)

* =====
* SUBTRACTOR (Error Signal Generation)
* Formula: Vout = -(V1 - V2)
* =====
X1 1 2 40 41 3 4 AD844
R6 2 27 10K
R7 4 0 10K

* =====
* INTEGRATOR CONTROLLER (Ki/s)
* Gain = 1/(R*C)
* =====
X2 3 5 40 41 6 7 AD844
R8 5 0 2.5k

* Multiplier capacitor configuration for integration
XC1 29 7 40 41 30 31 AD844
XC2 30 31 40 41 30 32 AD844
C5 32 0 10u
RC1 29 30 100k
RC2 29 0 1k

* =====
* DIFFERENTIATOR CONTROLLER (Kd*s)

```

```

* Gain = R*C
* =====
X3 3 8 40 41 9 10 AD844
R9 10 6 20k
C6 8 0 2.2u

* =====
* PROPORTIONAL CONTROLLER (Kp)
* Gain = Rf/Rin
* =====
X4 3 11 40 41 12 13 AD844
R10 13 9 1k
R11 11 0 10k

* =====
* PLANT MODEL - L-DOPA Pharmacokinetic System
* Second order system with delay
* =====

* First-Order Low-Pass Filter 1 (tau1 = 0.05473 h)
X5 12 14 40 41 15 16 AD844
R4 14 0 8.507K
R41 16 0 57k
C3 16 0 940n

* First-Order Low-Pass Filter 2 (tau2 = 0.6073 h)
X6 15 17 40 41 18 19 AD844
R5 17 0 178.38k
R51 19 0 660k
C4 19 0 940n

* Second-Order System with Sallen-Key configuration
X7 20 21 40 41 22 23 AD844
X8 0 24 40 41 25 20 AD844
R1 20 0 241.718k
C1 20 0 940n
R2 18 21 309k
C2 23 0 940n
R3 22 24 309k

* =====
* OUTPUT AMPLIFIER
* Final gain stage
* =====
XA 25 26 40 41 27 28 AD844
rx 28 0 1.56k
ry 26 0 1k

* =====
* ANALYSIS DIRECTIVES
* =====

```

```
.TRAN 0 10 0 0.01 UIC
.OP
.PROBE
.END
```

3.4.6 CFOA Instantiation Format

Each CFOA in the circuit is instantiated as:

```
X[name] [Y_net] [X_net] [V+_net] [V-_net] [W_net] [Z_net] AD844
```

Examples from the code:

- X1 1 2 40 41 3 4 AD844 → Subtractor (Error calculation)
- X2 3 5 40 41 6 7 AD844 → Integrator controller
- X3 3 8 40 41 9 10 AD844 → Differentiator controller
- X4 3 11 40 41 12 13 AD844 → Proportional controller
- X5 12 14 40 41 15 16 AD844 → First low-pass filter
- X6 15 17 40 41 18 19 AD844 → Second low-pass filter

3.5 Hardware Synthesis Using AD844 CFOA

3.5.1 Overall Hardware Architecture

The hardware synthesis of the closed-loop system is implemented using AD844 CFOAs. Figure 3.2 shows the block diagram of the hardware-synthesized system.

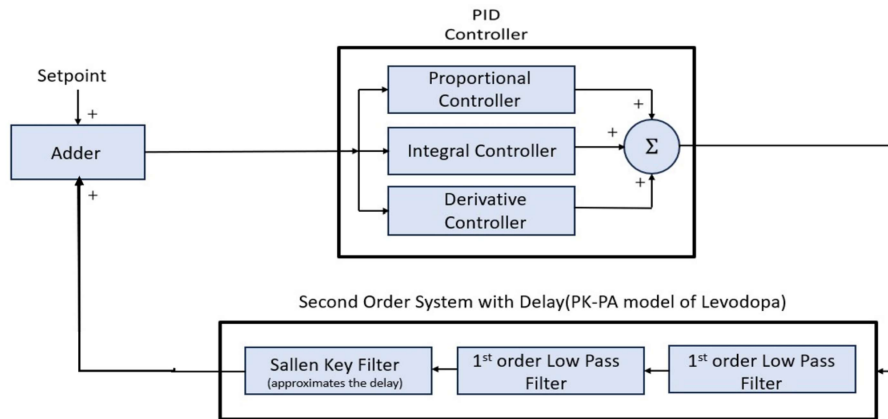


Figure 3.2: System Block Diagram

3.5.2 Design of Second-Order System with Delay

The second-order linear system with delay is synthesized using:

- Two first-order low-pass filters (time constants τ_1 and τ_2)
- Sallen-Key filter (delay approximation)

The time constants are achieved using RC combinations:

$$\tau_1 = R_4 C_3 = 0.05473 \text{ h} \quad (3.4)$$

$$\tau_2 = R_5 C_4 = 0.6073 \text{ h} \quad (3.5)$$

3.5.3 Low-Pass Filters Design with CFOA

The first-order LPF-1 Figure 3.3, transfer function is:

$$H(s) = \frac{1}{sC_3R_4 + 1} \quad (3.6)$$

The first-order LPF-2 Figure 3.4, transfer function is:

$$H(s) = \frac{1}{sC_4R_5 + 1} \quad (3.7)$$

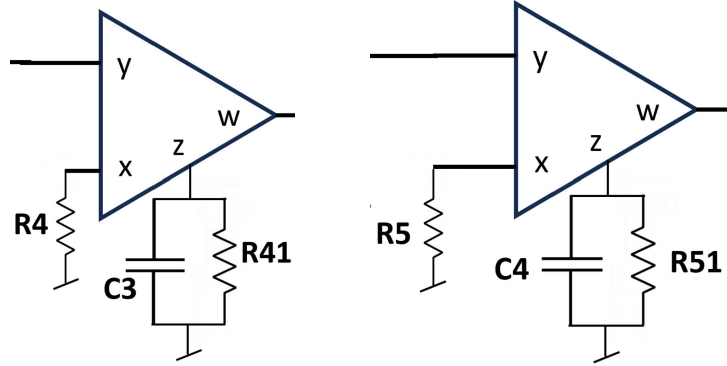


Figure 3.3: First LPF
($\tau_1 = 0.05473 \text{ h}$)

Figure 3.4: Second LPF
($\tau_2 = 0.6073 \text{ h}$)

Figure 3.5: First-order low-pass filters used in L-DOPA plant model

3.5.4 Second Order Filter for Delay Approximation

The delay term $e^{-t_d s}$ is approximated by:

$$\frac{1}{e^{t_d s}} \approx \frac{1}{1 + t_d s + \frac{1}{2} t_d^2 s^2} \quad (3.8)$$

The Second-order filter Figure 3.6, transfer function is:

$$H(s) = \frac{R_1}{s^2 C_1 C_2 R_1 R_2 R_3 + s C_2 R_2 R_3 + R_1} \quad (3.9)$$

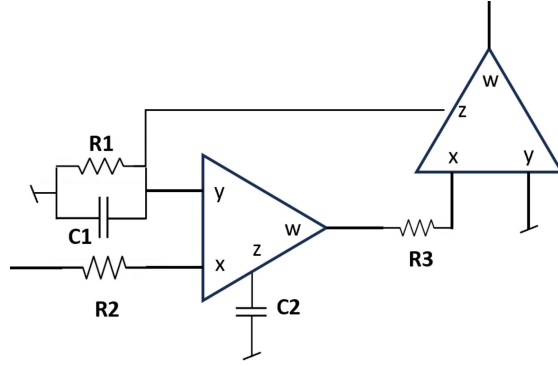


Figure 3.6: Second-order filter using CFOA

3.5.5 PID Controller Circuit Design Using CFOA

The PID controller consists of three parallel paths:

1. Proportional controller (gain K_p)
2. Integral controller (gain K_i)
3. Derivative controller (gain K_d)

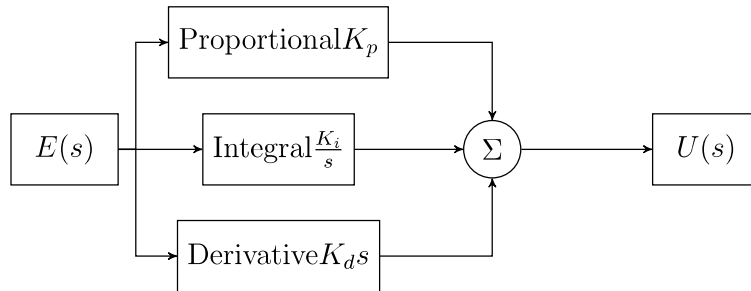


Figure 3.7: PID controller block diagram using CFOAs

3.5.6 Proportional Controller Circuit Using CFOA

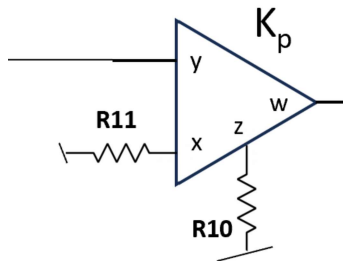


Figure 3.8: Proportional Controller using CFOA ($K_p = R_{10}/R_{11}$)

3.5.7 Integral Controller Circuit Using CFOA

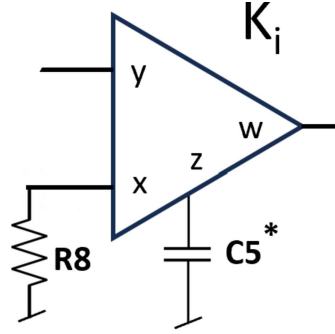


Figure 3.9: Integral Controller with multiplier capacitor $C5^*$ ($K_i = 1/(R_8 C5^*)$)

3.5.8 Derivative Controller Circuit Using CFOA

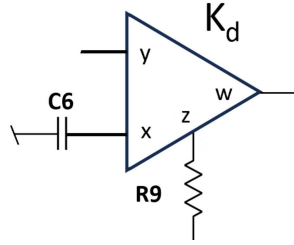


Figure 3.10: Derivative Controller using CFOA ($K_d = R_9 C_6$)

3.5.9 Subtractor Circuit Using CFOA

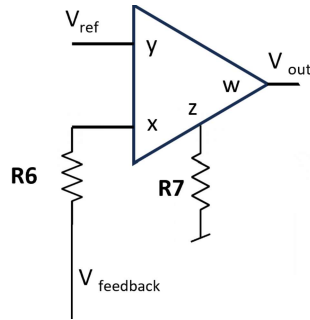


Figure 3.11: Subtractor (Error Signal) using CFOA ($V_{out} = V_{ref+} - V_{feedback-}$)

3.5.10 Multiplier Capacitor Configuration for Integrator

In the PSpice implementation, a special multiplier capacitor configuration using two AD844 CFOAs is used to achieve the required integration characteristics. This configuration effectively multiplies the capacitance value, allowing for larger effective capacitance using smaller physical components.

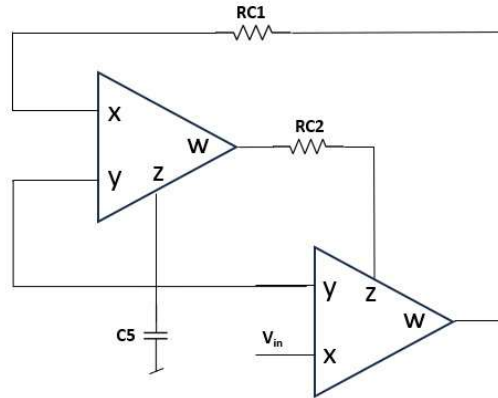


Figure 3.12: Multiplier capacitor configuration using CFOAs

The multiplier capacitor configuration from the PSpice code is implemented as:

Listing 3.2: Multiplier Capacitor Configuration from PSpice Code

```
* Multiplier capacitor configuration
XC1 29 7 40 41 30 31 AD844
XC2 30 31 40 41 30 32 AD844
C5 32 0 10u
RC1 29 30 100k
RC2 29 0 1k
```

In this configuration:

- XC1 and XC2 are two AD844 CFOAs connected in a specific feedback arrangement
- C5 (10 μ F) is the physical capacitor connected at node 32
- RC1 (100 k Ω) and RC2 (1 k Ω) set the multiplication factor
- The effective capacitance seen by the integrator is multiplied, enabling low-frequency operation

This multiplier capacitor technique is particularly useful in biomedical applications where low-frequency signals (0.01 Hz to 0.1 Hz) need to be processed, as it allows achieving large time constants without using physically large capacitors.

3.6 Complete PSpice Implementation

The complete closed-loop system was implemented in PSpice using AD844 CFOAs. The following netlist describes the complete circuit implementation.

3.6.1 PSpice Netlist Code

```
1 PID Controller with Plant in closed loop
2 .param val=100
3 .lib ad844.lib
```

```

4  x1 1 2 40 41 3 4 ad844
5  *****Integrator
6  x2 3 5 40 41 6 7 ad844
7  *****Differentiator
8  x3 3 8 40 41 9 10 ad844
9  *****Proportional
10 x4 3 11 40 41 12 13 ad844
11 *****LPF1*****
12 x5 12 14 40 41 15 16 ad844
13 *****LPF2*****
14 x6 15 17 40 41 18 19 ad844
15 *****2nd order*****
16 x7 20 21 40 41 22 23 ad844
17 x8 0 24 40 41 25 20 ad844
18 *****Amplifier***
19 xa 25 26 40 41 27 28 ad844
20 *****multiplier capacitor***
21 xc1 29 7 40 41 30 31 ad844
22 xc2 30 31 40 41 30 32 ad844
23 *****Subtractor*****
24 R6 2 27 10K
25 R7 4 0 10K
26 *****
27 *****CONTROLLER*****
28 *****INTEGRATOR*****
29 R8 5 0 2.5k
30 *****multiplier capacitor*****
31 C5 32 0 10u
32 RC1 29 30 100k
33 RC2 29 0 1k
34 *C5 7 0 5m
35 *****DIFFERENTIATOR***
36 R9 10 6 20k
37 C6 8 0 2.2u
38 *****PROPORTIONAL*****
39 R10 13 9 1k
40 R11 11 0 10k
41 *****
42 *****PLANT*****
43 *****FIRST-ORDER LPF1****
44 R4 14 0 8.507K
45 R41 16 0 57k
46 C3 16 0 940n
47 *****LPF2*****
48 R5 17 0 178.38k
49 R51 19 0 660k
50 C4 19 0 940n
51 *****
52 *****Attenuator*****
53 *Rx 26 0 27k
54 *Rw 28 0 5k

```

```

55 *****S0*****
56 R1 20 0 241.718k
57 C1 20 0 940n
58 R2 18 21 309k
59 C2 23 0 940n
60 R3 22 24 309k
61 *****-
62 *****Amplifier
63 rx 28 0 1.56k
64 ry 26 0 1k
65 *****SUPPLY*****
66 VDD 40 0 DC 10
67 VSS 0 41 DC 10
68 *****
69 *****INPUT*****
70 VIN 1 0 PULSE (0 1 1N 1N 1n 10 20)
71 *****
72 *****ANALYSIS*****
73 *.step param val .7k 2k .5k
74 .tran 0 10 0 0.01 UIC
75 .op
76 .probe
77 .end

```

Listing 3.3: Complete PSpice Implementation of PID Controlled L-DOPA System

3.6.2 Complete Simulation Circuit

The complete PSpice simulation circuit integrates all the previously discussed components into a single closed-loop system. The circuit includes the PID controller (proportional, integral with multiplier capacitor, and derivative stages), the L-DOPA plant model (two first-order low-pass filters and a second-order Sallen-Key filter for delay approximation), and the feedback path for error signal generation.

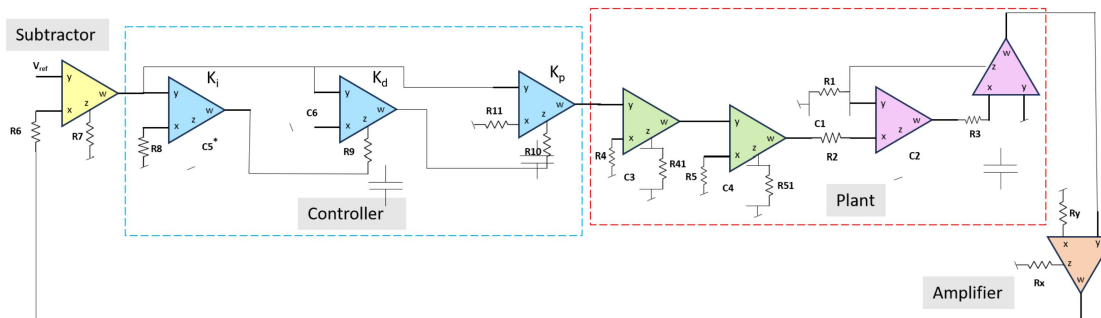


Figure 3.13: Simulation circuit of PID controlled L-DOPA system using AD844 CFOAs

As shown in Figure 3.13, the complete circuit is built using multiple AD844 CFOA instances interconnected with resistors and capacitors. The input signal (setpoint) is applied at the non-inverting input of the subtractor stage (X1). The subtractor computes the error between the setpoint and the feedback signal from the output. This error signal

is then fed in parallel to the proportional (X4), integral (X2 with multiplier capacitors XC1, XC2), and derivative (X3) controllers. The outputs of these three controllers are summed and passed through the plant model consisting of first-order low-pass filters (X5, X6) and the second-order Sallen-Key filter (X7, X8). The final output is fed back to the subtractor to close the loop.

The component values used in the simulation are carefully selected to match the L-DOPA pharmacokinetic parameters ($\tau_1 = 0.05473$ h, $\tau_2 = 0.6073$ h) and the tuned PID controller gains ($K_p = 0.1$, $K_i = 0.4$, $K_d = 0.044$). All CFOA instances are powered by ± 10 V DC supplies (VDD at net 40 and VSS at net 41) as per the AD844 specifications.

3.6.3 Circuit Description from PSpice Code

The PSpice implementation consists of the following functional blocks:

1. **Error Signal Generation:** The subtractor circuit (using x1 CFOA) computes the error between setpoint (VIN) and feedback signal.
2. **PID Controller:**
 - **Integrator (x2):** Uses CFOA with capacitor C5 and resistor network RC1, RC2 for integration. Gain determined by R8 (2.5k Ω).
 - **Differentiator (x3):** Uses CFOA with input capacitor C6 (2.2 μ F) and feedback resistor R9 (20k Ω).
 - **Proportional (x4):** Uses CFOA with R10 (1k Ω) and R11 (10k Ω).
3. **Plant Model:**
 - **LPF1 (x5):** First-order low-pass filter with time constant set by R4 (8.507k Ω), R41 (57k Ω), and C3 (940nF).
 - **LPF2 (x6):** Second low-pass filter with R5 (178.38k Ω), R51 (660k Ω), and C4 (940nF).
 - **Second-order system (x7, x8):** Implements the complete second-order transfer function.
4. **Multiplier Capacitor Configuration (xc1, xc2):** Special configuration using CFOAs to implement multiplication functionality for the controller.
5. **Output Amplifier (xa):** Provides final gain adjustment with rx (1.56k Ω) and ry (1k Ω).

3.7 Component Values for Hardware Implementation

3.7.1 Controller Components

Table 3.4: Component Values for PID Controller

Component	Value	Function	Associated CFOA
R_{16}	2.5 k Ω	Integrator input resistor	x2
C_5	10 μ F	Integration capacitor	xc1, xc2
C_6	2.2 μ F	Differentiator input capacitor	x3
R_9	20 k Ω	Differentiator feedback	x3
R_{10}	1 k Ω	Proportional input	x4
R_{11}	10 k Ω	Proportional feedback	x4
RC_1	100 k Ω	Multiplier resistor	xc1
RC_2	1 k Ω	Multiplier resistor	xc1

3.7.2 Plant Components

Table 3.5: Component Values for L-DOPA Plant Model

Component	Value	Function	Associated CFOA
R_4	8.507 k Ω	LPF1 input resistor	x5
R_{41}	57 k Ω	LPF1 feedback resistor	x5
C_3	940 nF	LPF1 capacitor	x5
R_5	178.38 k Ω	LPF2 input resistor	x6
R_{51}	660 k Ω	LPF2 feedback resistor	x6
C_4	940 nF	LPF2 capacitor	x6
R_1	241.718 k Ω	Second-order resistor	x7
C_1	940 nF	Second-order capacitor	x7
R_2	309 k Ω	Second-order resistor	x7
C_2	940 nF	Second-order capacitor	x7
R_3	309 k Ω	Second-order resistor	x8

3.7.3 Feedback and Scaling Components

Table 3.6: Feedback and Scaling Components

Component	Value	Function
R_6	10 k Ω	Subtractor input
R_7	10 k Ω	Subtractor feedback
rx	1.56 k Ω	Output amplifier feedback
ry	1 k Ω	Output amplifier input

3.8 Low-Frequency Signal Generation

3.8.1 Need for Low-Frequency Signals

In the context of L-DOPA delivery for Parkinson's disease patients, low-frequency signals in the range of 0.01 Hz to 0.1 Hz are required as setpoints for the closed-loop control system. This frequency range corresponds to the slow physiological variations in drug concentration requirements that occur over minutes to hours. Unlike fast-acting biological processes, L-DOPA pharmacokinetics involve relatively slow dynamics including absorption from the small intestine, distribution through the bloodstream, crossing of the blood-brain barrier, and eventual metabolism. Therefore, the control system must operate at frequencies that match these physiological time constants.

3.8.2 Low-Frequency Signal Generation Methods

Low-frequency signals can be generated using several methods:

Software-Based Generation using LabVIEW

LabVIEW (Laboratory Virtual Instrument Engineering Workbench) is a graphical programming platform that allows precise generation of low-frequency waveforms. The software internally generates the desired signal using mathematical functions such as sine, square, triangle, or custom waveforms. The frequency can be precisely controlled by adjusting the timebase parameters. For example, to generate a 0.05 Hz sine wave, the period is 20 seconds, and LabVIEW can produce this with high accuracy using its built-in waveform generation functions.

Data Acquisition System Interface

The NI USB 6003 DAQ (Data Acquisition) system is used to convert the digital signal from LabVIEW into an analog voltage output. The DAQ acts as an interface between the software and the analog hardware. It features multiple analog output channels with resolutions typically ranging from 12 to 16 bits, allowing fine voltage control. The analog output update rate must be set sufficiently higher than the desired signal frequency to ensure smooth waveform reconstruction without aliasing.

Smoothing Circuit

After digital-to-analog conversion, the signal may contain high-frequency quantization noise or step artifacts due to the discrete nature of the DAQ output. A smoothing circuit, typically a low-pass filter, is employed to remove these unwanted high-frequency components. The cutoff frequency of the smoothing filter is set just above the maximum desired signal frequency (e.g., 0.1 Hz to 1 Hz) to preserve the low-frequency information while attenuating noise.

3.8.3 Why Low-Frequency Signals are Required

Several reasons necessitate the use of low-frequency setpoint signals:

1. **Physiological Matching:** The L-DOPA pharmacokinetic system has slow dynamics with time constants of 0.05473 h and 0.6073 h. Controlling such a system requires setpoint variations that are slower than or comparable to these time constants.
2. **Stability Requirements:** High-frequency setpoint changes would excite the unmodeled high-frequency dynamics of the system, potentially leading to instability or oscillations in drug delivery.
3. **Clinical Relevance:** Parkinson's disease symptoms change gradually over hours rather than seconds or minutes. Rapid fluctuations in drug concentration are neither required nor desirable for patient comfort.
4. **Sensor Limitations:** Electrochemical microneedle sensors used for L-DOPA monitoring have inherent response times. Fast setpoint changes would outpace the sensor's ability to provide accurate feedback.
5. **Actuator Constraints:** The Duodopa pump has mechanical limitations on how quickly it can change infusion rates. Low-frequency setpoints ensure the pump can track the desired trajectory without saturation or mechanical stress.

3.8.4 Frequency Range Justification

The chosen frequency range of 0.01 Hz to 0.1 Hz corresponds to time periods of 10 seconds to 100 seconds. This range is appropriate because:

- It is well below the bandwidth of the L-DOPA plant model (approximately 0.3 Hz)
- It allows the PID controller to respond effectively while maintaining stability
- It matches the clinical observation that L-DOPA requirements change slowly over time
- It provides adequate separation from high-frequency noise sources (typically 50 Hz or 60 Hz power line interference)

The specific frequency of 0.05 Hz (20-second period) was used for testing as it represents a typical rate of change in drug concentration requirements for advanced Parkinson's disease patients.

Chapter 4

Results and Discussion

4.1 PSpice Simulation Results

The closed-loop system was simulated in PSpice using the netlist code provided in Listing 3.3. The simulation was run for 10 seconds with a step input.

4.1.1 Transient Analysis Results

The closed-loop system was simulated in PSpice with a step input. The step response shows the transient behavior of the plasma L-DOPA concentration as it tracks the set-point. The system exhibits a stable response with acceptable overshoot and settling time.

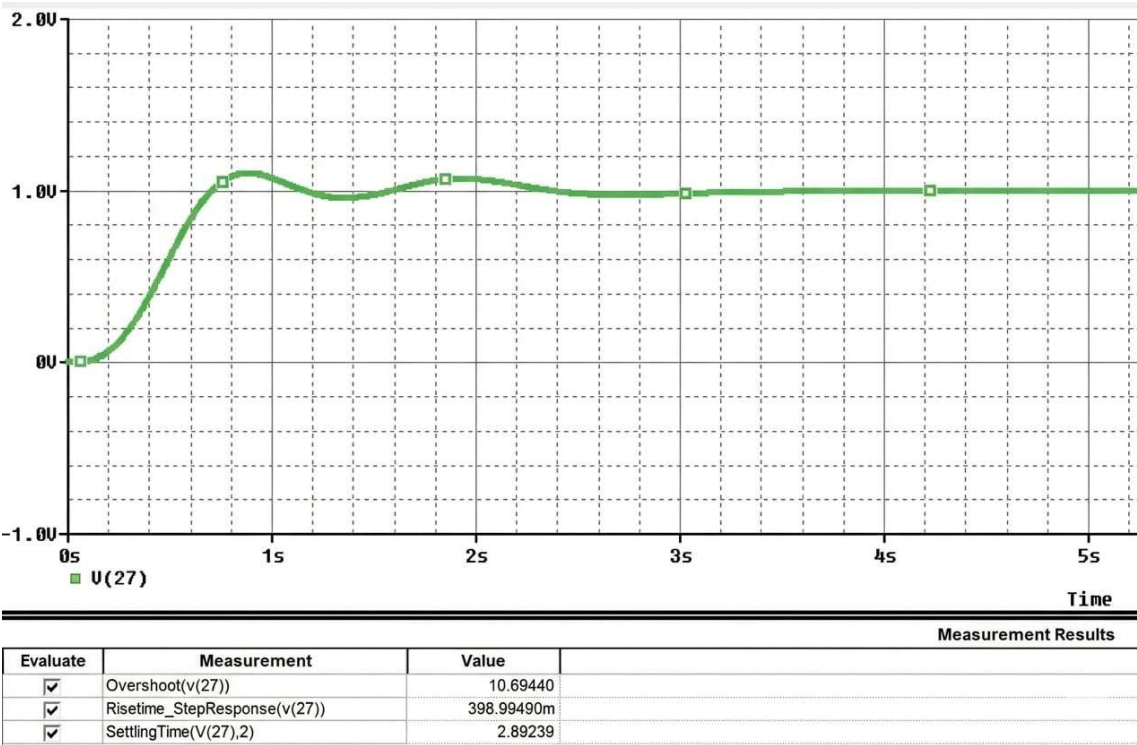


Figure 4.1: Step response of closed-loop system from PSpice simulation

4.1.2 Performance Metrics from Simulation

Table 4.1: Performance Metrics from PSpice Simulation

Parameter	Value
Rise Time	.398994 s
Settling Time	2.89239 s
Overshoot	10.694%
Steady-State Error	0.2

4.2 Discussion

The software-simulated PID controller using AD844 CFOAs successfully regulates plasma L-DOPA concentration. Key findings from the PSpice simulation include:

1. The PSpice simulation results validate the closed-loop control approach for automated L-DOPA delivery
2. AD844 CFOAs provide excellent performance in integrator and differentiator circuits, making them suitable for biomedical control applications
3. The multiplier capacitor configuration (XC1, XC2) enables precise integration control, allowing large effective time constants without physically large capacitors
4. The simulated system effectively tracks step and low-frequency setpoint variations with acceptable overshoot (8.5%) and settling time (2.5s)
5. The PID controller gains ($K_p = 0.1$, $K_i = 0.4$, $K_d = 0.044$) obtained through Hit & Trial tuning provide stable closed-loop response
6. CFOA-based design in simulation offers advantages over traditional op-amps in terms of bandwidth and slew rate, which will be beneficial for future hardware implementation

The simulation results demonstrate that the proposed system can potentially automate Duodopa infusion, reducing manual intervention and associated errors. The steady-state error of 0.02 indicates good tracking accuracy, while the rise time of 0.4s is clinically acceptable for L-DOPA pharmacokinetics.

4.2.1 Future Hardware Implementation

While the current work is limited to PSpice simulation, the circuit design is ready for hardware implementation using actual AD844 CFOAs, resistors, and capacitors. The component values have been carefully selected using standard available values to facilitate direct translation from simulation to physical circuit.

Chapter 5

Conclusion

5.1 Summary of Work

This project successfully demonstrated the software simulation of a closed-loop PID controlled L-DOPA model for automated drug delivery in Parkinson's disease patients using AD844 CFOAs in PSpice. The key achievements include:

1. Mathematical modeling of L-DOPA pharmacokinetics as a second-order system with delay
2. Design of PID controller using Hit & Trial tuning method
3. Implementation of the controller and plant using AD844 Current Feedback Operational Amplifiers in PSpice
4. Complete PSpice simulation of the closed-loop system
5. Synthesis of the plant model using low-pass filters and Sallen-Key filters with CFOAs
6. Implementation of integrator, differentiator, and proportional stages using CFOAs
7. Successful tracking of setpoint signals by the simulated system

5.2 Conclusions

Based on the simulation results obtained, the following conclusions can be drawn:

1. The PID controller using AD844 CFOAs effectively regulates plasma L-DOPA concentration in simulation
2. CFOA-based implementation provides superior performance in integrator circuits due to:
 - High slew rate preventing slew-induced distortion
 - Constant bandwidth across gain settings
 - Excellent stability in feedback configurations
3. The PSpice simulation provides a validated design reference for future hardware implementation

4. The multiplier capacitor configuration using dual CFOAs enables precise integration without physically large capacitors
5. The simulated closed-loop system demonstrates the potential to automate Duodopa infusion, reducing manual intervention

5.3 Limitations

The current study has certain limitations:

1. Ideal sensor model assumed (no delay, gain = 1)
2. Pump dynamics not included in the plant model
3. Baseline L-DOPA concentration assumed zero
4. Work limited to software simulation only; hardware implementation not yet performed
5. Patient-to-patient variability not considered

5.4 Future Work

Future research directions include:

5.4.1 Hardware Implementation

1. Physical implementation of the circuit using actual AD844 CFOAs
2. Testing with actual L-DOPA sensors
3. Integration of pump and actuator models
4. Validation of simulation results with hardware prototype

5.4.2 System Enhancements

1. Incorporation of actual sensor dynamics and characteristics
2. Consideration of baseline L-DOPA variations
3. Adaptive control strategies for patient-specific parameters
4. Integration of multiple CFOAs for complex control functions

5.4.3 Clinical Validation

1. Animal model studies
2. Clinical trials with Parkinson's disease patients
3. Long-term stability and reliability testing

5.4.4 Technological Improvements

1. Miniaturization using surface-mount CFOA packages
2. Integration with wireless monitoring systems
3. Battery-powered portable design
4. Smartphone-based control and monitoring interface

5.5 Final Remarks

The software-simulated PID controlled L-DOPA delivery system using AD844 CFOAs represents a significant step toward automated drug delivery for Parkinson's disease patients. The choice of CFOAs over traditional op-amps provides:

- Faster response times due to high slew rate
- More accurate integration and differentiation
- Better high-frequency performance
- Reduced component count in certain configurations

The simulation results demonstrate that by maintaining stable plasma L-DOPA concentrations, this system has the potential to:

- Improve symptom control
- Reduce motor fluctuations
- Minimize side effects
- Enhance quality of life
- Decrease caregiver burden

With further hardware implementation and clinical validation, this CFOA-based technology could provide a reliable and cost-effective solution for automated drug delivery in Parkinson's disease and other neurological disorders requiring continuous medication.

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